GroupDRO with Heterogeneous Feature Spaces

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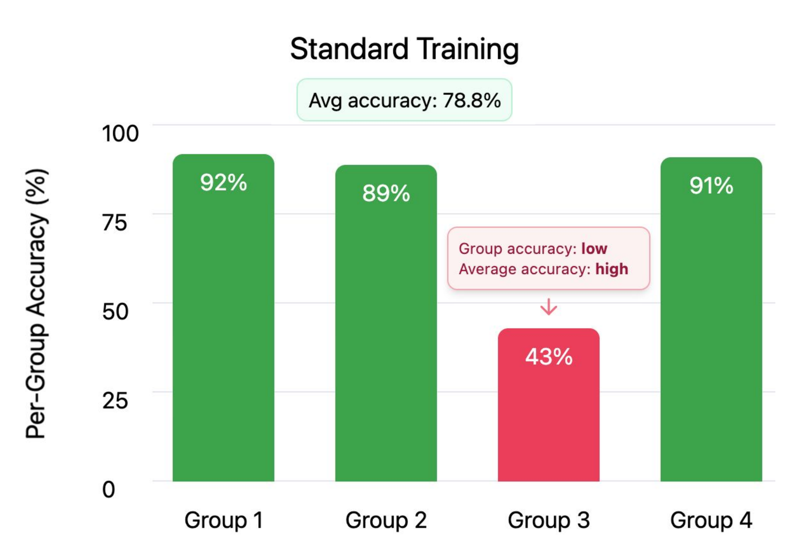
Abstract

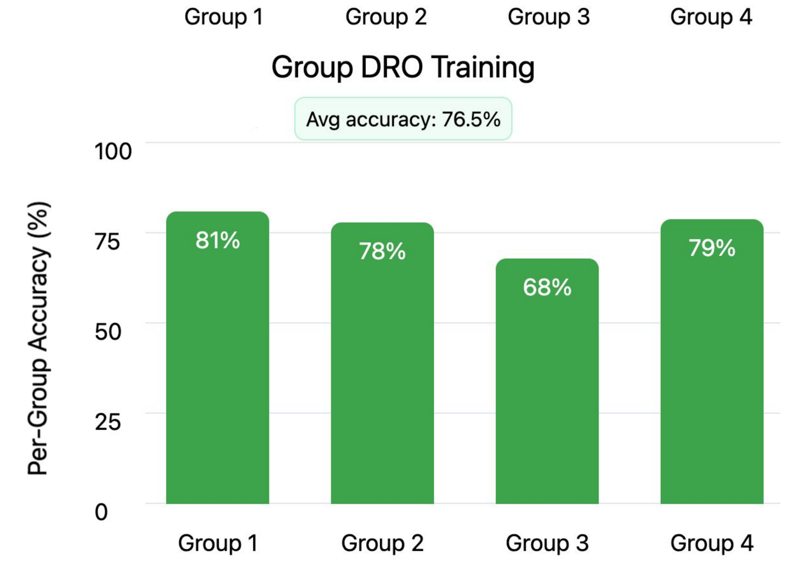
Modern models are increasingly deployed across heterogeneous input sources, where each group has its own feature space. Standard empirical risk minimization and Group Distributionally Robust Optimization (GroupDRO) usually assume a shared feature representation and fail to protect minority groups when feature spaces differ or some groups have far fewer samples. We study a supervised multi group setting with one encoder per group that maps its features into a shared latent space and a single classifier on top. To align groups in this space, we give each class c a Gaussian anchor and penalize the 2-Wasserstein distance between its anchor and the corresponding empirical Gaussians, while also training on synthetic samples from the anchors to ensure classes are sufficiently separated. We extend this framework with a GroupDRO objective that minimizes the worst-case group loss. Experiments on MNIST/USPS (heterogeneous image resolutions), UCI Fed Heart Disease (heterogeneous clinical features across 4 hospitals), and NHANES CVD (heterogeneous feature availability across 3 assessment groups) show that per-group encoders combined with Gaussian anchors and GroupDRO improve worst-group accuracy by up to +15% over standard baselines.

Motivation

Problem 1: Minority groups and spurious correlations

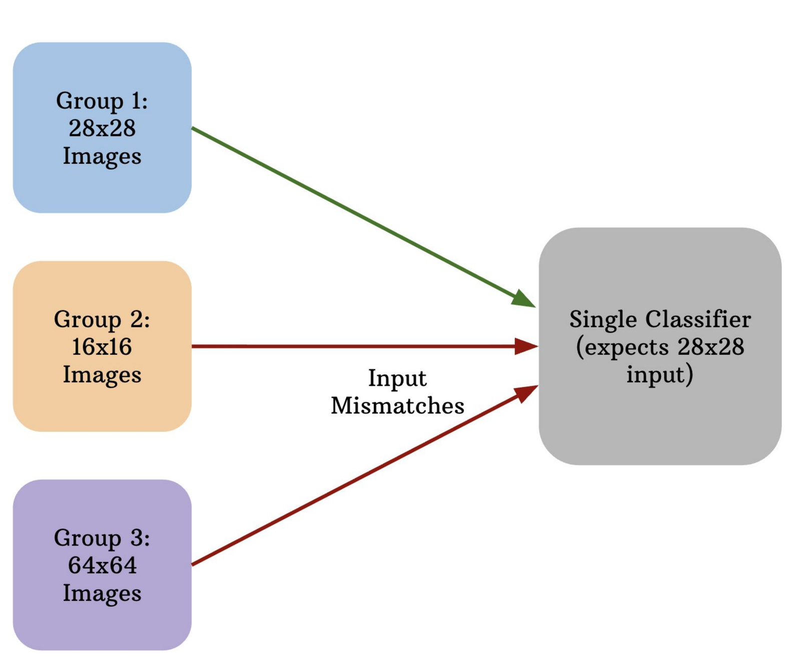
When data quality or quantity differs across groups, standard training often favors the majority, leading to high average accuracy but poor performance and spurious correlations for minority groups. In practice, we want models that keep performance high for every group so that behavior is robust and reliable, not just good on average.





Problem 2: Heterogeneous feature spaces

In many applications, groups live in different input spaces, such as images with different resolutions or medical records with different recorded features. Most GroupDRO methods assume a single shared input representation, which fails when groups live in different feature spaces.



Representation Learning Architecture

Model: Group-specific Encoders and Shared Classifiers

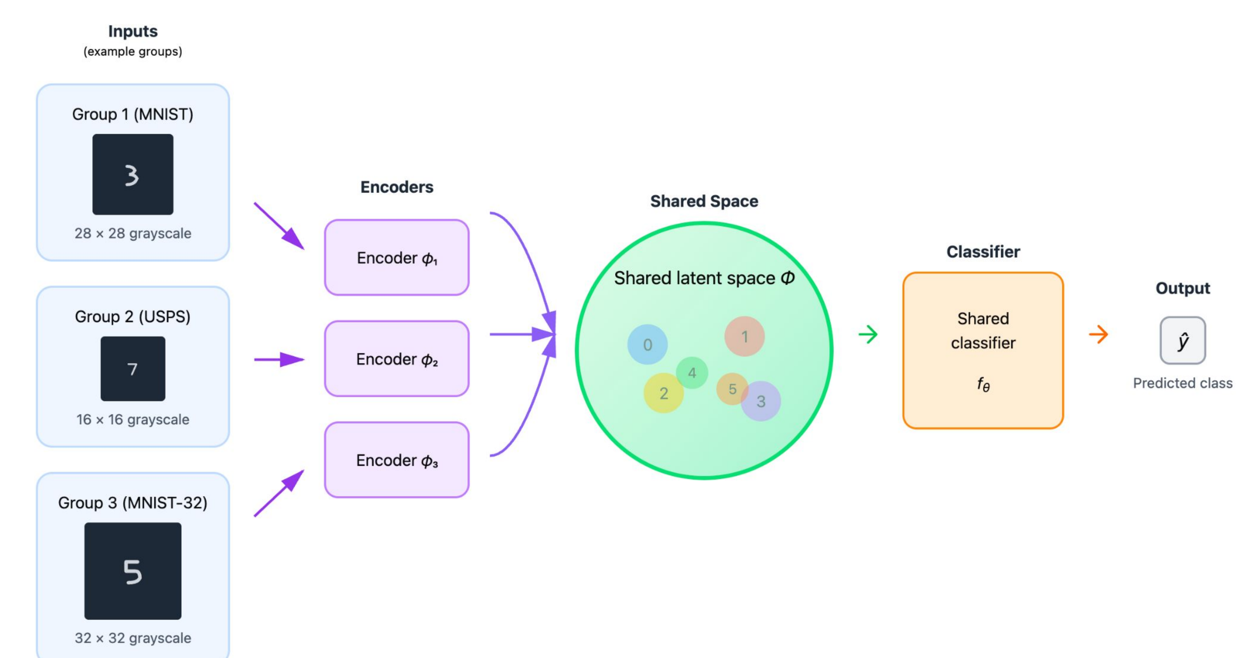
We work in a centralized setting with data (x_i, y_i, g_i), where each group g has its own input space X_g and potentially different dimensionalities. For each group we learn a separate encoder ϕ_g that maps inputs from X_g into a shared latent space Φ of fixed dimension. A single classifier f_θ then operates on this latent representation to produce class probabilities.

Model Prediction for (*x*, *g*):

$$\hat{p}(y \mid x_i, g_i) = f_\theta(\phi_{g_i}(x_i))$$

Learnable Encoder Function:

$$\phi_g : \mathcal{X}_g \to \Phi \subset \mathbb{R}^k$$



Latent Class Anchors for Cross-group Alignment

To align how different groups represent the same class in latent space, we give each class c a learnable Gaussian anchor μ_c. For each class we fit an empirical Gaussian to its latent embeddings (aggregated across groups) and penalize its squared 2-Wasserstein distance to μ_c, encouraging all groups to map the same class to a common region of the latent space. We also draw synthetic latent points from each anchor and train the classifier on them, which keeps anchors for different classes separated and prevents collapse.

Anchor Fitting Loss

$$L_{\text{fit}} = \sum_{c=1}^{C} W_2^2\big(\mathcal{N}(\hat{m}_c, \hat{S}_c), \mathcal{N}(m_c, S_c)\big)$$

Anchor Separation Loss

$$L_{\text{sep}} = \mathbb{E}_c\, \mathbb{E}_{z \sim \mathcal{N}(m_c, S_c)}\big[\ell(f_\theta(z), c)\big]$$

GroupDRO Objective

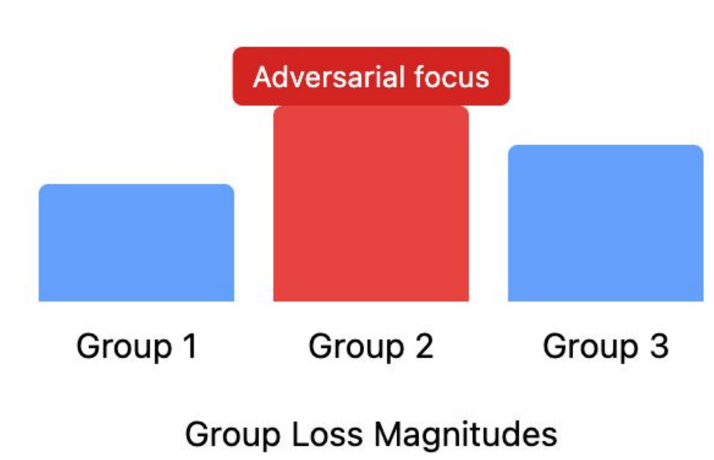
For each group g we define a group loss $L^{(g)}$ that combines three components: a supervised classification loss on that group, an anchor fitting loss for its embeddings, and the shared anchor separation loss. Instead of minimizing the simple average of these group losses, GroupDRO introduces a weight vector q over groups and and optimizes a min–max objective to minimizes the worst weighted combination, focusing on the hardest groups.

Per Group Loss:

$$L^{(g)} = L^{(g)}_{\text{clf}} + \lambda_{\text{fit}} L^{(g)}_{\text{fit}} + \lambda_{\text{sep}} L_{\text{sep}}$$

GroupDRO min-max objective:

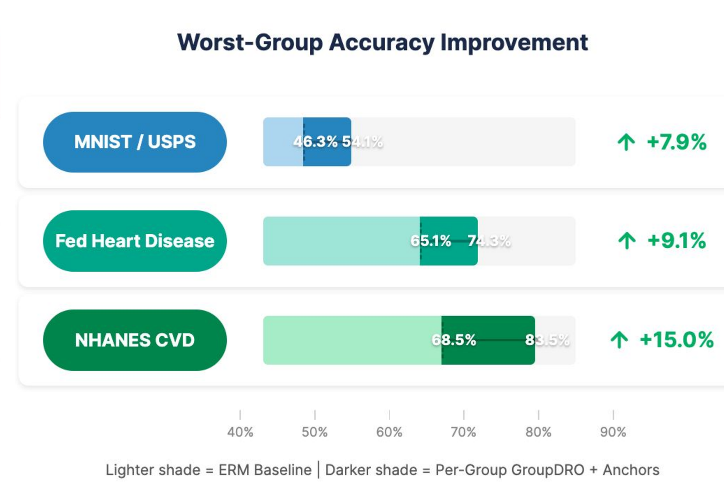
$$\min_{\theta, \{\phi_g\}, \{\mu_c\}} \max_{q \in \Delta^G} \sum_{g=1}^{G} q_g\, L^{(g)}$$



Experimental Results

We evaluate our framework across three datasets with increasing complexity. MNIST/USPS tests heterogeneous image resolutions. Fed Heart Disease tests heterogeneous clinical features across 4 hospitals. NHANES CVD tests natural feature-availability heterogeneity with cardiovascular disease questionnaire data. For each dataset, we compare ERM baselines against our full method (per-group encoders + Gaussian anchors + GroupDRO). Results are averaged over 5-10 random seeds.

Dataset	Task	Groups	Group Details	Heterogeneity Type	Samples
MNIST/USPS	Digit classification (10 classes)	2	G0: MNIST (28×28 images), G1: USPS (16×16 images)	Different image resolutions	13,000
Fed Heart Disease	Heart disease detection (binary)	4	G0: Cleveland (13f), G1: Hungarian (12f), G2: Switzerland (11f), G3: VA (12f)	Different clinical features per hospital	740
NHANES CVD	CVD prediction (binary)	3	G0: Survey only (15f), G1: Survey+Exam (18f), G2: Survey+Exam+Labs+BP (25f)	Different feature availability	17,005



Fed Heart Disease — Per-Group Accuracy (10 seeds)

Each hospital has different available clinical features (11-13 features)

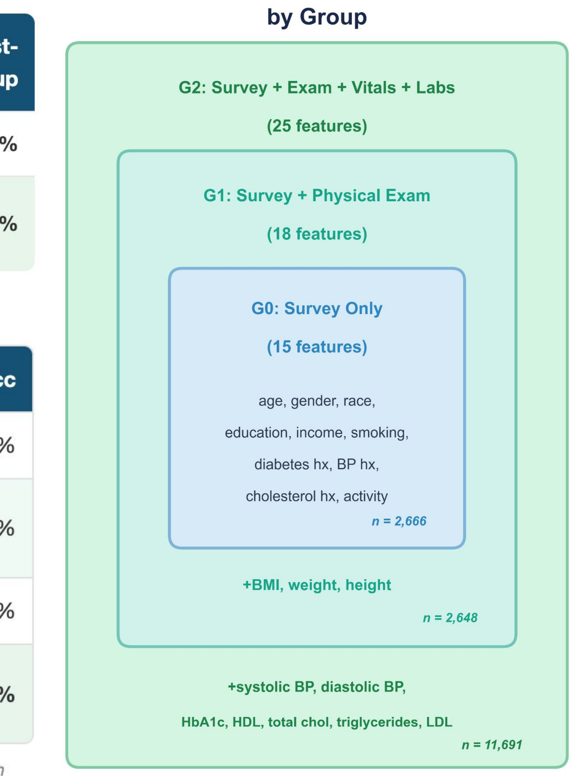
Config	G0 Cleveland (13f)	G1 Hungarian (12f)	G2 Switzerland (11f)	G3 VA (12f)	Worst-Group
Shared ERM	75.3%	76.9%	68.8%	65.6%	**65.1%**
Per-group GroupDRO	74.5%	75.2%	**93.8%**	78.0%	**74.3%**

NHANES CVD — Main Results (5 seeds)

Method	Worst-Group Acc	Overall Acc	Balanced Acc
Common-features ERM (15f)	68.50 ± 1.96%	69.51 ± 1.93%	70.58 ± 1.87%
Common-features GroupDRO (15f)	69.07 ± 1.14%	70.04 ± 1.20%	71.06 ± 1.47%
Per-group ERM (15/18/25f)	75.27 ± 1.97%	80.67 ± 3.85%	79.62 ± 3.12%
Per-group GroupDRO (15/18/25f)	**83.52 ± 1.72%**	**84.99 ± 2.03%**	**85.48 ± 1.23%**

Rows 1-2 use standard architecture (2-layer MLP, 64-dim latent) on 15 common features. Rows 3-4 use deep architecture (4-layer MLP, 128-dim latent) with per-group feature sets.

NHANES Feature Availability by Group



Conclusion

Our framework uses group-specific encoders mapping heterogeneous feature spaces into a shared latent space, aligns classes with Gaussian anchors via 2-Wasserstein distance, and trains with a GroupDRO objective that focuses on worst-performing groups. Across MNIST/USPS, Fed Heart Disease, and NHANES CVD, per-group encoders with Gaussian anchors and GroupDRO improve worst-group accuracy by +7.9% to +15.0% over ERM baselines. A key finding is that anchors and GroupDRO are synergistic: neither helps much alone, but combined they produce large gains by structuring the shared latent space so GroupDRO can effectively reweight across heterogeneous groups. Future work includes extending to additional datasets, theoretical convergence guarantees, and adapting the framework to federated learning settings.

[1] Rakotomamonjy, Alain, et al. "Personalised federated learning on heterogeneous feature spaces." arXiv preprint arXiv:2301.11447 (2023).

[2] Ogier du Terrail, Jean, et al. "FLamby: Datasets and Benchmarks for Cross-Silo Federated Learning in Realistic Healthcare Settings." Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track (2022)

[3] Sagawa, S., et al. "Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization." ICLR (2020).

Collaborated with Xenia Konti and Michael Zavlanos